



Handwriting Recognition

Image Processing and Internship

210090 – Bekhruz Pulatov

Presentation Outline

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Problem Statement and Motivation

- **Poor Quality of Handwritten Images**
- **Inaccuracy in Recognizing Handwritten Text**
- **Inefficiency in Manual Document Digitization**

Project Objectives

- **Apply image processing techniques to preprocess handwritten text images.**
- **To design and develop a system that can recognize handwritten English characters and words accurately.**
- **To demonstrate practical applications such as document digitization and automated text input.**

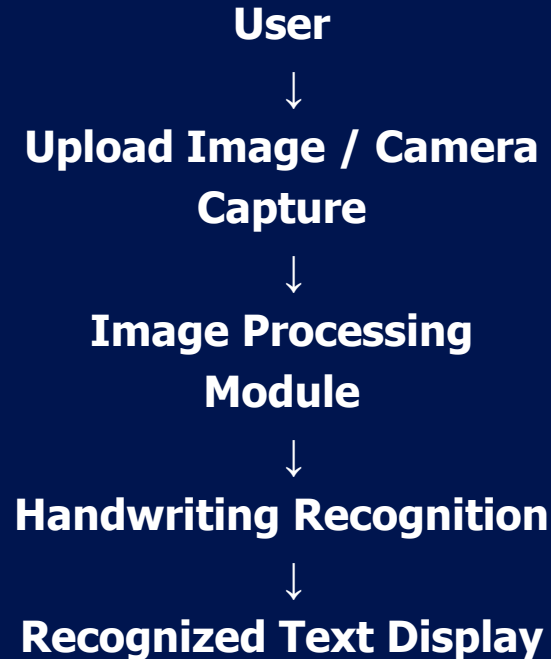
Project Timeline

Phase	Activity	Duration
Phase 1	Literature review and requirement analysis	1 week
Phase 2	Data collection and preprocessing	2 weeks
Phase 3	Model design and training	3 weeks
Phase 4	Testing and evaluation	1 week
Phase 5	System implementation and documentation	1 week
Total Duration		9 Weeks

Solutions and Systems Architecture

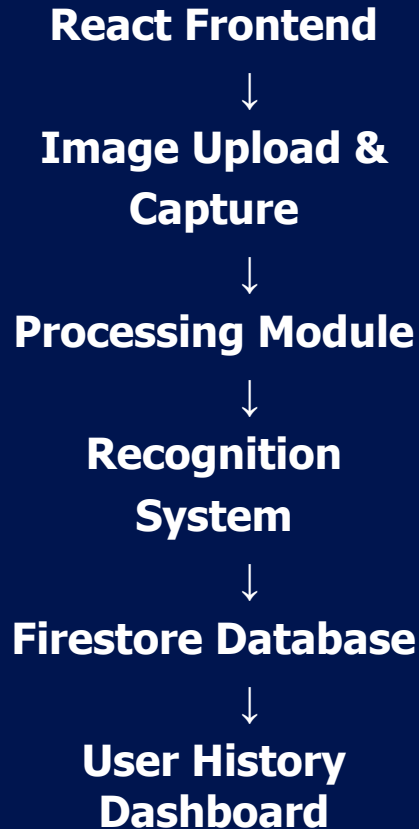
User Interaction Flow - Part 1

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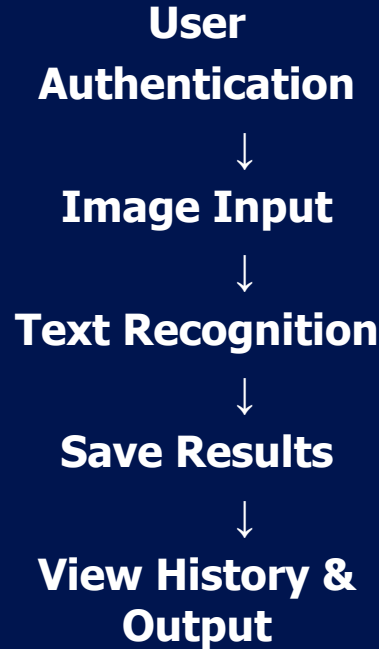
System Component Interaction - Part 2

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Application Workflow - Part 3

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Tools and Technologies Used

React & Tailwind CSS - Part 1

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Technology	Purpose
React	To build the user interface
Tailwind CSS	For responsive UI design

GitHub, Typescript & OpenCV - Part 2

Technology	Purpose
GitHub	For project management and code storage
TypeScript	For structured and safer coding
OpenCV	For preprocessing handwritten images

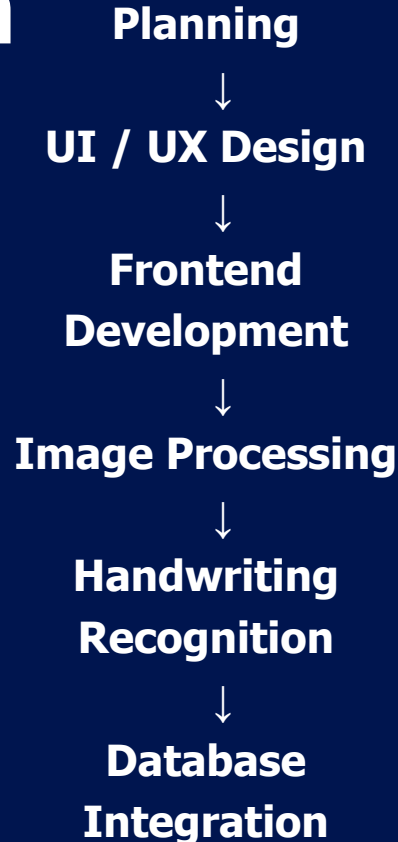
Firestore Storage & Firebase

Authentication - Part 3

Technology	Purpose
Firestore Storage	Stores user history and recognized text
Firebase Authentication	Handles user login and authentication

Development Process

Development Process Diagram



Challenges Encountered and Resolutions

Challenges

- **Variability in Handwriting Styles**
- **Noise and Distortions**
- **Character Ambiguity**

Resolutions

- **Develop a robust recognition model**
- **Improve image preprocessing techniques**
- **Increase character recognition accuracy by distinguishing between visually similar characters**

Results

Result 1 - Recognition Performance

- **Successfully recognized handwritten English text from uploaded images.**
- **Processed handwritten input through image upload and camera capture features.**
- **Generated editable digital text output with improved readability.**

Result 2 - System Functionality Results

- **Implemented secure user authentication and login system.**
- **Stored recognition history using Firestore database integration.**
- **Enabled users to view, manage and delete previous recognition results.**

Result 3 - Challenge Resolution Results

- **Reduced image noise using preprocessing techniques.**
- **Improved recognition accuracy for different handwriting styles.**
- **Achieved faster processing and smoother user interaction through optimized workflow.**

Future Scope and Impact

Future Scope

- **AI Accuracy Improvement**
- **Signature Verification Feature**
- **Offline Recognition Mode**

Impact of the Project

- **Faster Document Digitization**
- **Educational and Business Benefits**
- **Support for Digital Transformation**

Summary and Conclusion

Summary and Conclusion

In conclusion, an English Handwriting Recognition System is a powerful application of Artificial Intelligence that aims to convert handwritten text into digital form, despite challenges such as varying writing styles and noise. By using modern system techniques, we can achieve high accuracy and efficiency. Overall, these systems play an important role in digitization, automation and accessibility, and will continue to improve as technology advances.

Project Contribution Breakdown

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Student ID	Student Name	Roles and Responsibilities	Specific Tasks Completed	Key outputs / Results
210090	Bekhruz	Backend Development	Authentication Setup, Database Integration, Text Recognition Processing	Secure login system, Data and history storage, Digital text conversion
210090	Bekhruz	Coordinator	Project management, Team coordination, Testing supervision	Organized workflow, Smooth integration, Improved quality
210090	Bekhruz	Frontend Development	UI Development, Image Upload & Camera Capture, Result Display	User - friendly interface, Handwritten image input, Recognized text output

References

- 1 Plamondon, R., and Srihari, S. N. (2000). On-line and off-line handwriting recognition: A comprehensive survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 22(1), 63–84.
- 2 LeCun, Y., Bottou, L., Bengio, Y., and Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11), 2278–2324.
- 3 Graves, A., Fernández, S., Gomez, F., and Schmidhuber, J. (2009). Connectionist temporal classification: Labelling unsegmented sequence data. *Proceedings of the 26th International Conference on Machine Learning*.
- 4 Marti, U. V., and Bunke, H. (2002). The IAM-database: An English sentence database for offline handwriting recognition. *International Journal on Document Analysis and Recognition*, 5(1), 39–46.

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**Thank you for your
attention!**